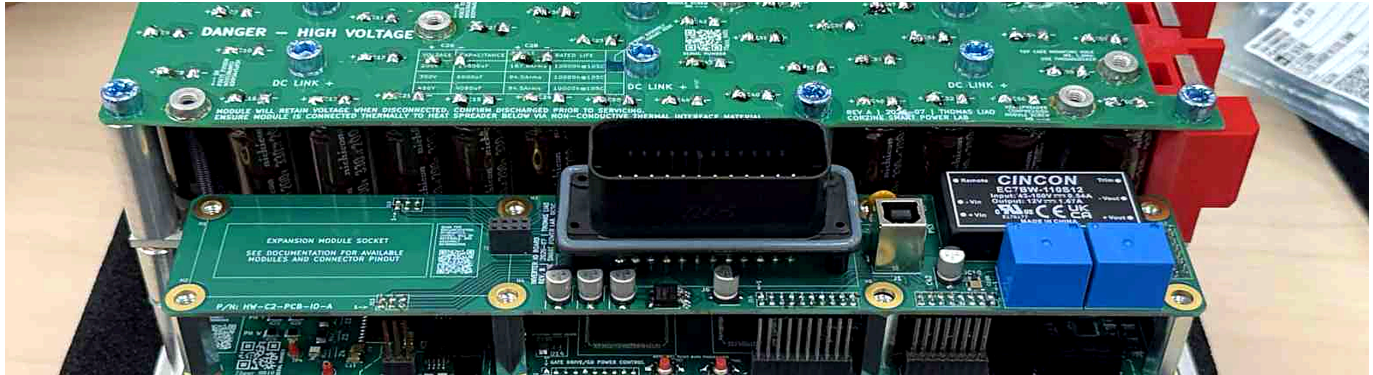


The open, safety-engineered power-conversion platform

Built on our bench-tested dual-MCU control architecture. Swappable power stages scale from 10 kW to the megawatt class for motor drives and DC/DC conversion, with SiC-ready designs.



Size 2 is a 200–450 V-class design rated 465 A continuous and 600 A peak for 60 seconds, with self-commissioning for PMSM and induction motors.

■ Built and bench-tested

Size 2 has completed a **180 V bring-up**, a **one-hour endurance run**, a **20-minute reversal stress test**, and instrument-verified motor self-commissioning. The Size 2 chassis is being tested for use on a **2018 Zero DSR electric motorcycle**.

■ Safety engineered

Six independent paths to the safe state combine hardware shutdown with main-controller and safety-coprocessor authority. The HARA and FMEA map to **ISO 26262** and **IEC 61800-5-2**, with hazards traced to safety requirements and validation.

■ Open everything

Schematics, firmware, **RTE Studio**, safety analyses, assembly data, and interactive telemetry are public at docs.openvvf.org. Sponsors and researchers can inspect what was built and how it performed.

MODULAR POWER ARCHITECTURE

Reusable control, scalable power stages

CONTROL MODULE

Dual-MCU control · FOC · independent safety

POWER STAGE

Gate-driver boards · capacitor boards · IGBT or SiC modules



Size 1

Approximately 10 kW class for light EVs and scalable designs

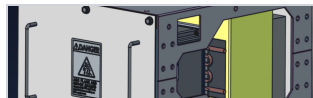
■ **EARLY DESIGN**



Size 2 inverter

200–450 V class · 465 A continuous design rating

■ **IMPLEMENTED, UNDER TEST**



Size 2 DC/DC

15 kW · 450 V class · reuses Size 2 boards and power stage

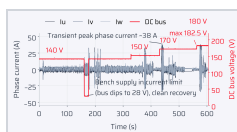
■ **RELEASED DESIGN**



Size 3

Up to 1,200 V · 1,400 A · modular liquid-cooled enclosure

■ **IN DEVELOPMENT**



Live telemetry

Ten published reports include raw, interactive logs for the browser-based Telemetry Viewer. Approximately 38 A peak current and clean recovery from a supply current-limit event.



SUPPORT THE SMART POWER LAB

Smart Power Lab contacts

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CURRENT SUPPORTERS

